



DroneX AI

MACHINE LEARNING ALGORITHMS

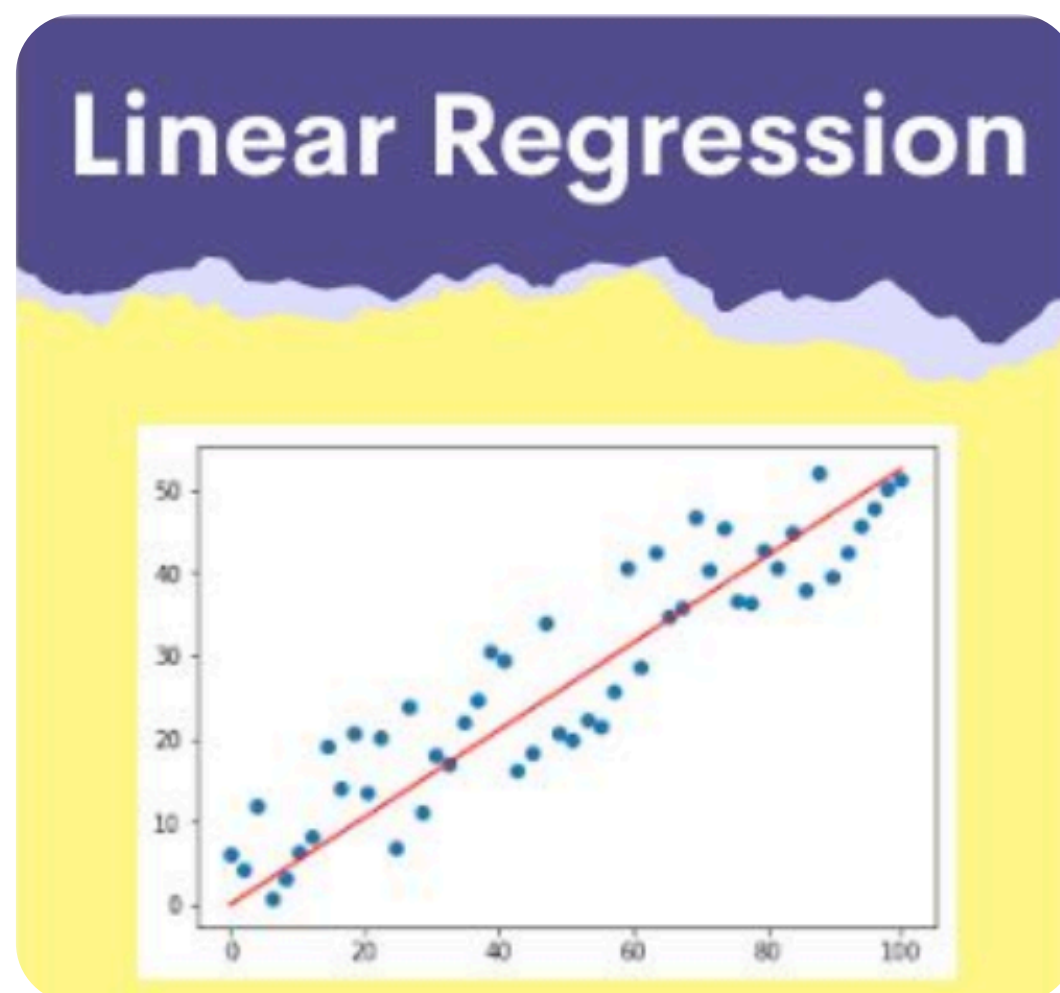


Karn Singh



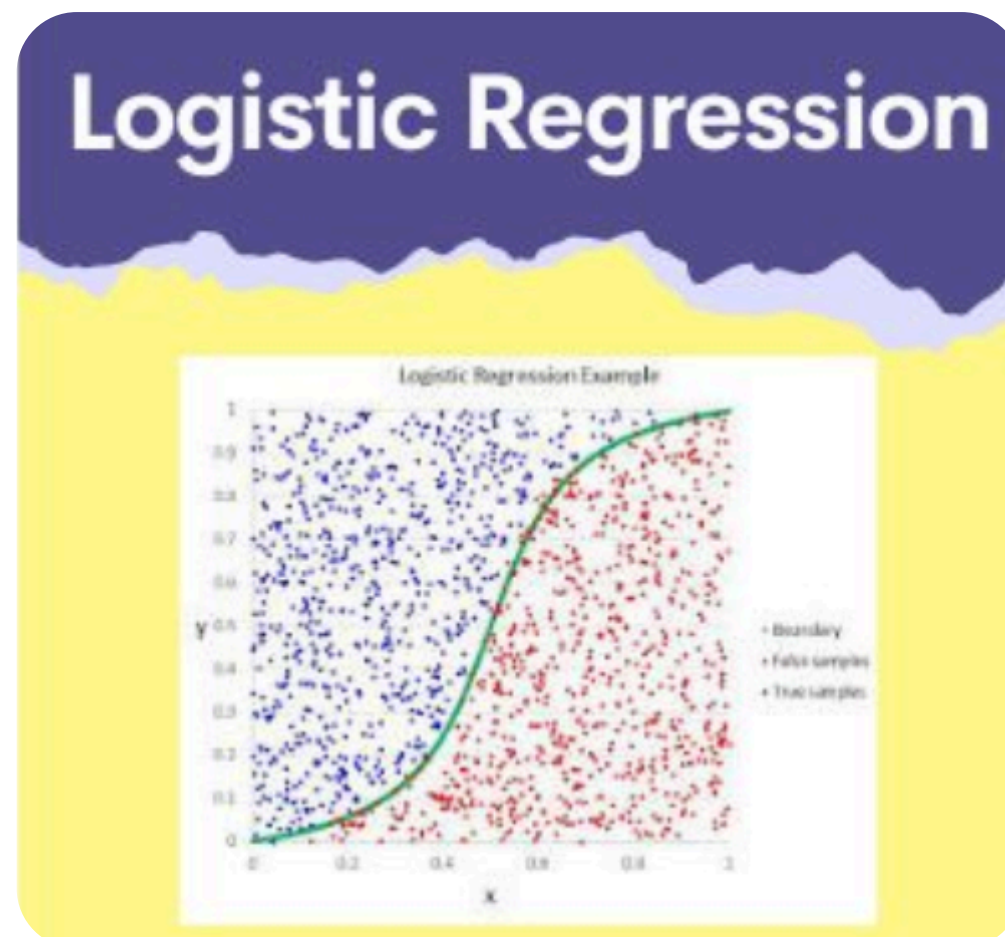
Linear Regression

- A statistical method used to model the relationship between independent (input) and dependent (output) variables.
- Best suited for predicting **continuous numerical outcomes** based on linear relationships.
- The model fits a **straight line** (regression line) that minimizes the error between predicted and actual values.
- **Example:** Predicting house prices, sales forecasts, or exam scores based on input factors.



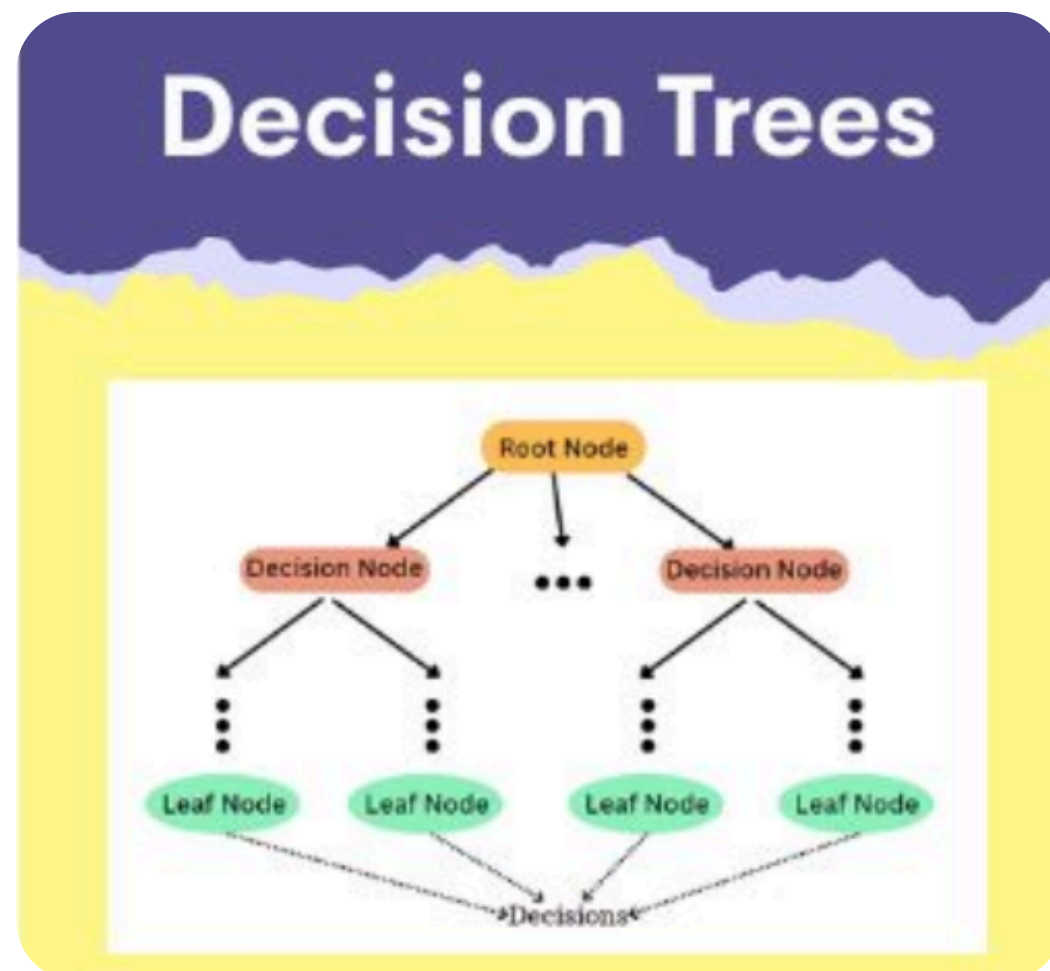
Logistic Regression

- A statistical method used for **classification tasks** where the output is categorical (e.g., Yes/No, 0/1, True/False).
- It estimates the **probability** that a given input belongs to a certain class using the **sigmoid function**.
- Outputs are mapped between **0 and 1**, making it suitable for binary and multi-class classification.
- **Example:** Spam detection (spam or not spam), disease diagnosis (positive/negative), customer churn prediction (leave/stay).



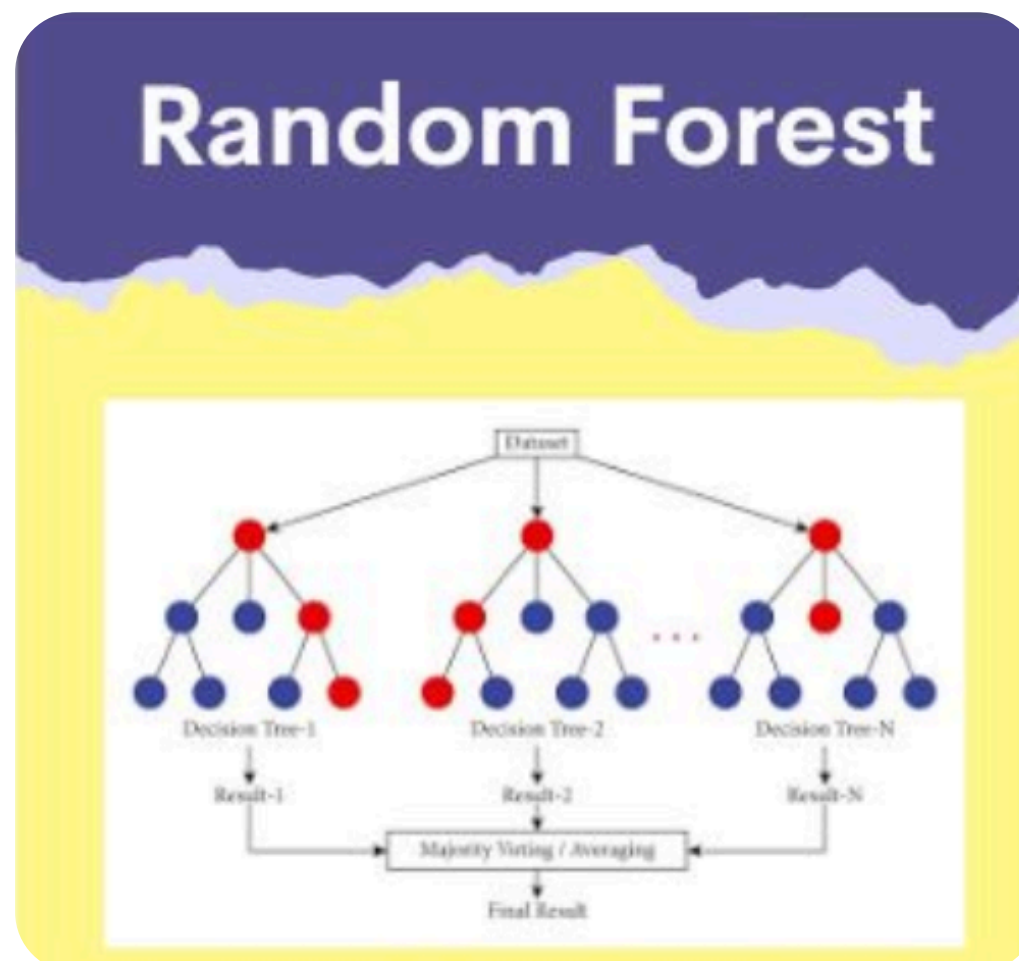
Decision Trees

- A supervised learning algorithm that splits data into branches based on **decision rules**.
- Works for both **classification** (categorical outcomes) and **regression** (numerical outcomes).
- Structure includes:
 - **Root Node**: Starting point of decisions.
 - **Decision Nodes**: Points where conditions split data.
 - **Leaf Nodes**: Final outcomes or predictions.
- **Example**: Predicting loan approval (Yes/No), diagnosing diseases, customer segmentation, or sales forecasting.



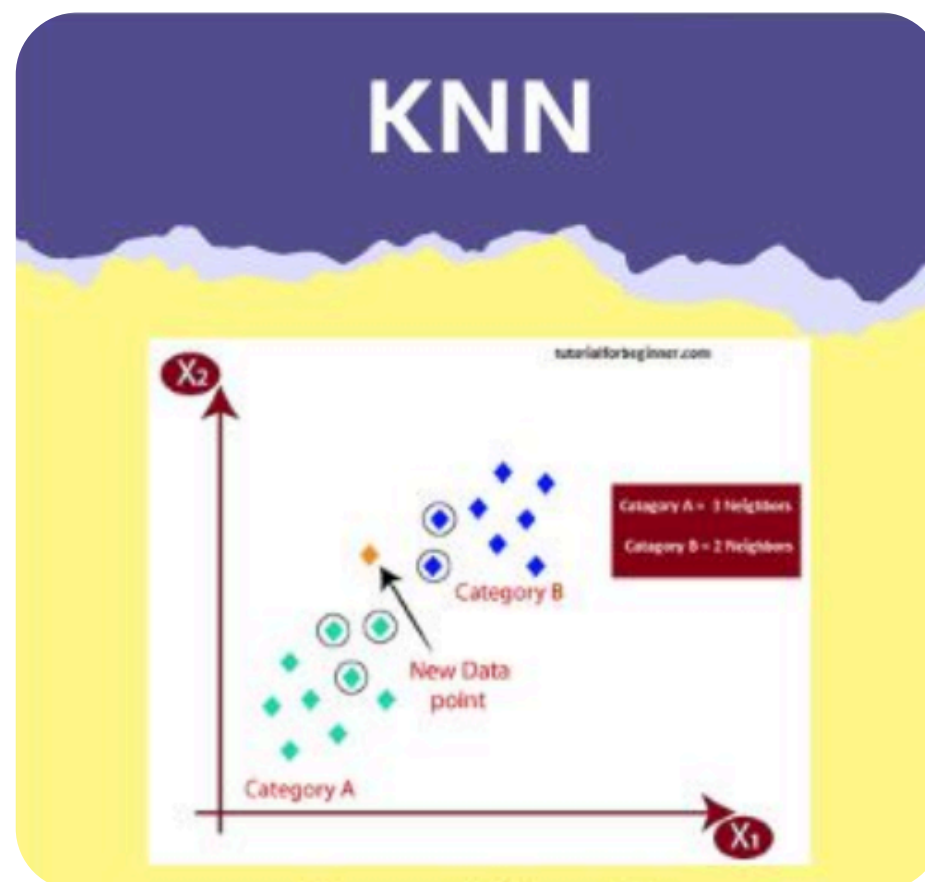
Random Forest

- An **ensemble learning method** that combines multiple decision trees to improve accuracy and reduce overfitting.
- Each tree is trained on a random subset of data and features.
- Final prediction is made by:
 - **Majority voting** (for classification)
 - **Averaging** (for regression)
- **Example:** Fraud detection, medical diagnosis, stock market prediction, customer churn analysis.



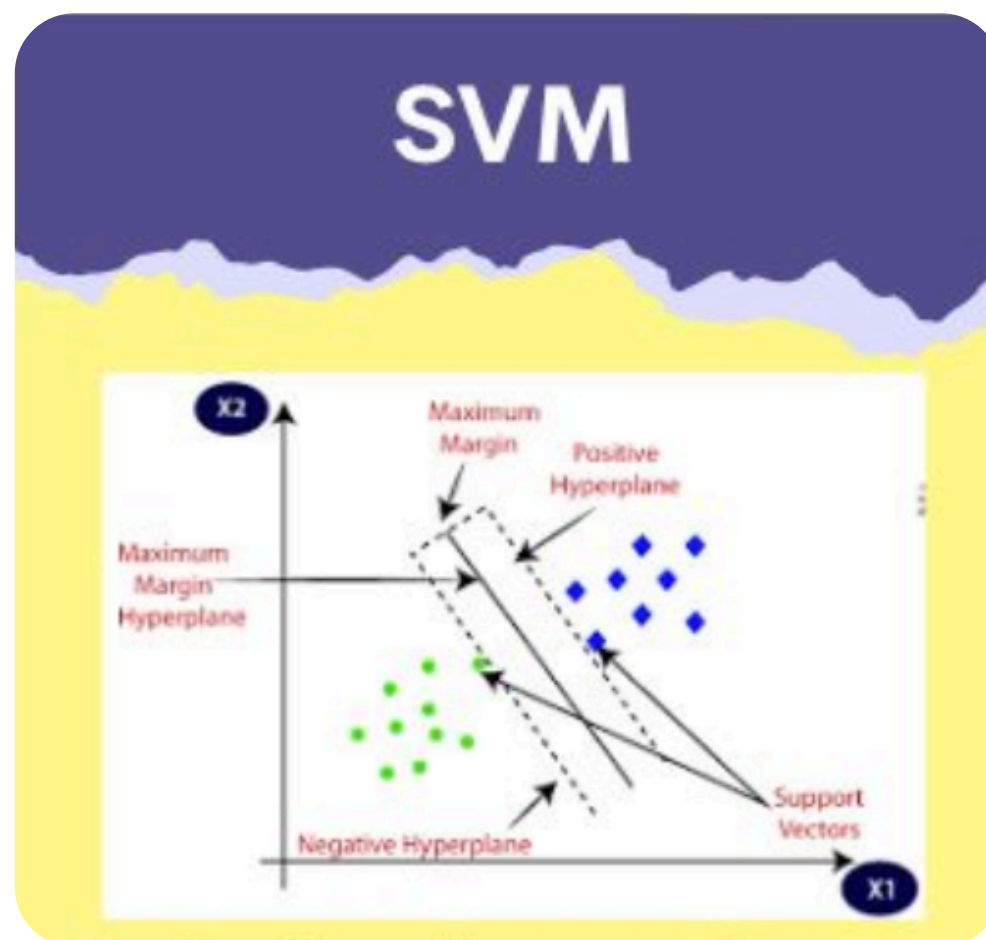
K-Nearest Neighbors (KNN)

- A simple, non-parametric algorithm used for **classification and regression**.
- Works by comparing the **distance** between a new data point and existing data points.
- The new data point is assigned to the **majority class** among its k nearest neighbors.
- Choice of k (number of neighbors) affects accuracy and performance.
- **Example:** Handwriting recognition, recommendation systems, medical diagnosis, and pattern detection.



Support Vector Machine (SVM)

- A powerful supervised learning algorithm used for classification and regression tasks.
- Works by finding the optimal hyperplane that best separates data points of different classes.
- Maximizes the margin (distance) between the hyperplane and the nearest data points (called support vectors).
- Can handle linear and non-linear classification using kernel functions.
- Example: Face recognition, text categorization, spam detection, and bioinformatics (disease classification).

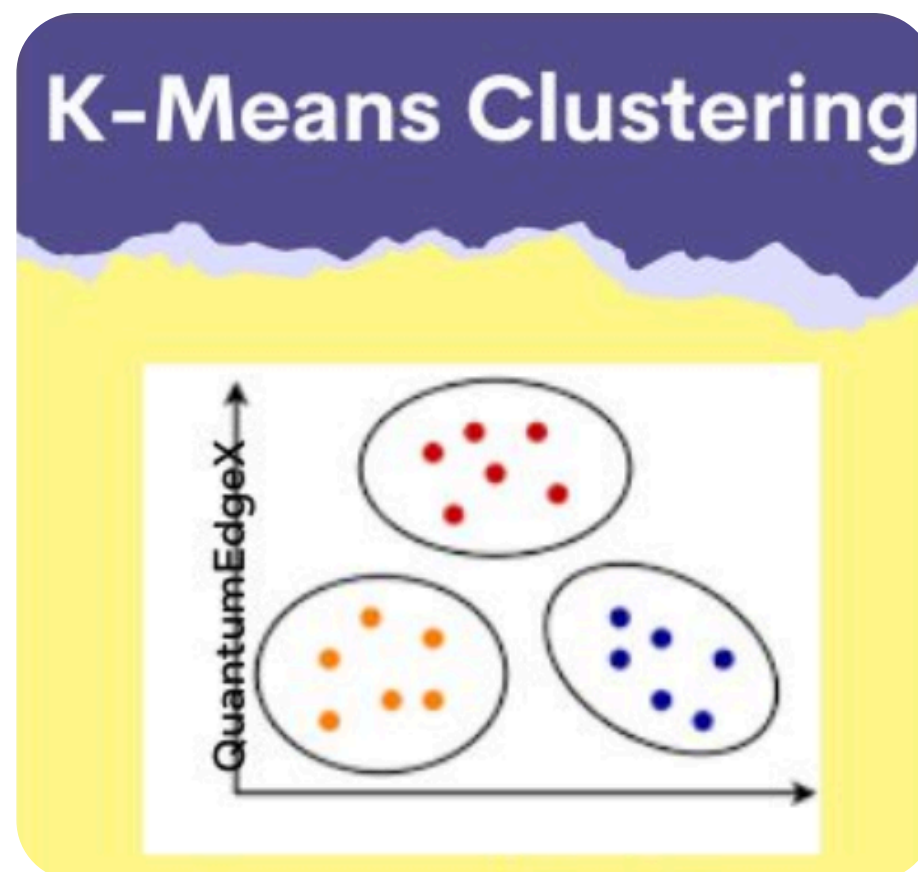


K-Means Clustering

- An **unsupervised learning algorithm** used to group data points into K clusters based on similarity.

Works by:

- Randomly selecting **K centroids**.
- Assigning each data point to the nearest centroid.
- Updating centroids until cluster assignments stabilize.
- Finds hidden **patterns or groupings** without prior labels.
- **Example:** Customer segmentation, market basket analysis, image compression, document clustering.

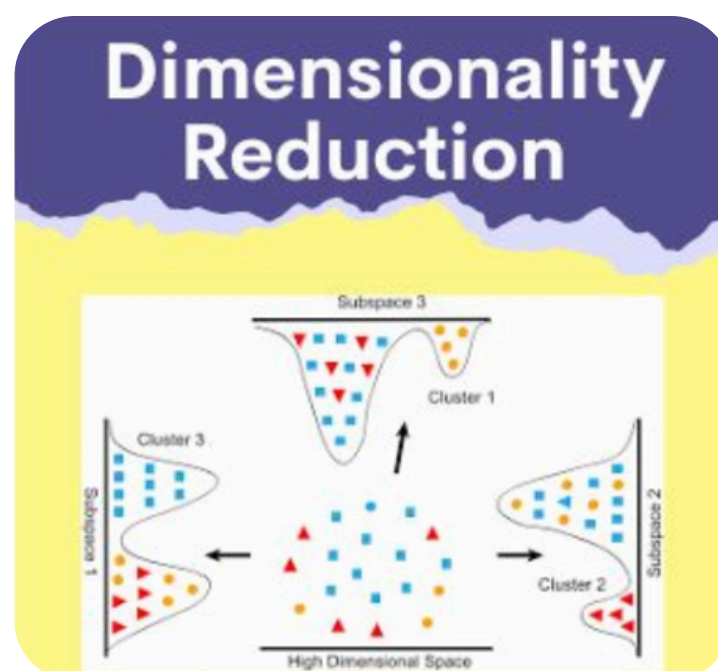


Dimensionality Reduction

- A technique used to reduce the number of **features (dimensions)** in a dataset while retaining essential information.
- Helps in **removing noise**, avoiding overfitting, and making models **faster and more efficient**.

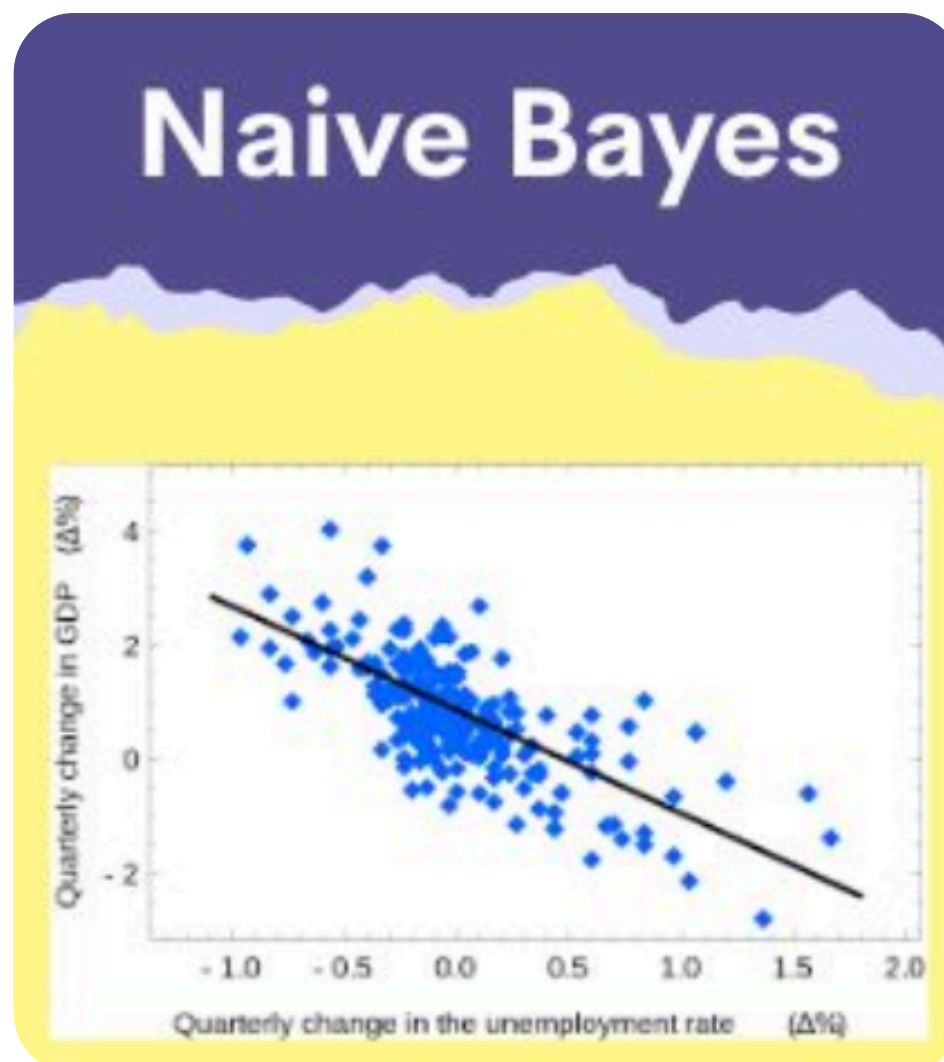
Common methods:

- **PCA (Principal Component Analysis)**
- **t-SNE (t-distributed Stochastic Neighbor Embedding)**
- **LDA (Linear Discriminant Analysis)**
- Especially useful for **high-dimensional data** like images, text, and genomics.
- **Example:** Image compression, text classification, face recognition, and visualization of large datasets.



Naive Bayes Classifier

- A **probabilistic machine learning algorithm** based on **Bayes' Theorem**.
- Assumes that features are **independent** (the “naive” assumption).
- Works well for **classification tasks** with large datasets.
- Particularly effective for **text classification** and spam filtering.
- **Example:** Email spam detection, sentiment analysis, document categorization, and medical diagnosis.





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